Attention Mechanisms in Self-Improving AI Systems: A Position Paper Bridging Continual Learning and Recursive Self-Improvement

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Abstract

Two research streams that emerged independently in 2024–2026 are converging on a shared technical substrate. Self-distillation fine-tuning (SDFT) [1], [3] addresses catastrophic forgetting during continual learning by using a model's own in-context-learned outputs as preservation targets, allowing new skill acquisition while keeping previously acquired capabilities intact. Recursive language models (RLMs) [7], [8], recursive self-improvement (RSI) systems [12], and the broader inference-time scaling toolkit [13], [14] address open-ended capability extension by letting models manage, refine, or rewrite their own context, prompt, or scaffold across iterations. On the surface these are distinct problems. We argue, however, that they share a common substrate: the *attention dynamics* that govern which tokens, sub-models, or steps a model relies on at each point in its computation. Continual learning succeeds when general-purpose attention pathways survive task-specific fine-tuning. Recursive self-improvement succeeds when attention either remains stable across improvement iterations or modifies in directions that demonstrably improve task performance. In both cases, what separates a working loop from a degenerate one is whether attention patterns remain interpretable, controllable, and approximately conserved.

This position paper synthesizes the SDFT, RLM, RSI, and inference-time-scaling literatures through the lens of attention dynamics. We propose a five-metric framework, the *Attention Stability Index* (ASI), comprising attention pattern correlation (APC), head activation entropy (HAE), cross-task attention transfer (CATT), capability preservation index (CPI), and head importance drift (HID). We give precise definitions, computation procedures, and empirical baselines drawn from public model releases. We then articulate four open problems that the unification surfaces: the relationship between attention stability and reasoning emergence; the role of attention-aware loss terms in continual learning; the design of attention-stable RSI loops; and the interpretability challenges of cross-model attention in deep recursive systems. We close with a 2026–2030 research roadmap and a discussion of governance and safety considerations specific to systems whose attention dynamics evolve at deployment time. We make no novel-method or experimental contribution; rather, we contribute a unifying conceptual frame, a measurement vocabulary, and an open-problems agenda.

Index Terms

Attention mechanisms, large language models, continual learning, self-distillation, recursive self-improvement, recursive language models, inference-time scaling, position paper.

I. Introduction

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THE dominant capability gains in large language models since 2022 have come from three distinct axes: pretraining scale [17], [18], post-training methods (RLHF, DPO, supervised fine-tuning), and inference-time scaffolds [7], [13], [14]. The third axis, inference-time scaffolds, has received concentrated attention since OpenAI o1's release in late 2024 and DeepSeek-R1's open-weights replication in January 2025 [14], [15]. Concurrent with the inference-scaling work, two adjacent communities have made progress on related problems. The first is continual learning: the long-standing challenge of acquiring new skills without losing old ones, now sharply revived for LLMs by self-distillation fine-tuning (SDFT) [1], [3], which avoids replay buffers and external regularizers by using the model's own in-context-learned outputs as preservation targets. The second is recursive self-improvement (RSI), which has crossed from theoretical prediction into deployed practice; the ICLR 2026 Workshop on AI with Recursive Self-Improvement accepted 110 papers, spanning self-play, automated AI research, prompt rewriting, and weight-level continual fine-tuning [12].

These three streams, inference-scaling, SDFT-style continual learning, and RSI, are usually discussed in separate venues, by distinct subcommunities, with different vocabularies. We argue that they share a deep technical substrate: the *attention dynamics* of the underlying transformer [16]. Specifically:

- In SDFT, what determines whether catastrophic forgetting is avoided is whether attention pathways for the prior tasks remain intact after fine-tuning on the new task. The mechanism by which SDFT succeeds, generating distilled targets that bridge the prior and new distributions, is fundamentally about *not perturbing* the attention patterns that produced the prior capabilities. We document this with reference to recent mechanistic-interpretability work on attention head activation patterns [28], [29].
- In RLM [7] and context folding [9], what determines whether the system is more accurate than direct prompting is whether the parent model's attention to sub-model outputs is appropriately calibrated. The parent model is no longer attending only to tokens; it is attending to entire sub-trajectories, summaries, and tool-call results. A working RLM is one whose cross-model attention is stable across task instances; a failing one is one where the parent loses calibration on the sub-output's reliability.
- In RSI loops, what determines whether the loop "actually gets better" (the operative question of the ICLR 2026 RSI Workshop [12]) is whether the agent's attention patterns evolve in directions that improve task performance, rather than collapsing to degenerate fixed points or oscillating between plausible but useless variants. The N2M-RSI conjecture [34] explicitly frames this as an information-theoretic threshold question: any scalable RSI implementation either throttles improvement or crosses an information-integration threshold in $O(\log T)$ time, with attention dynamics as the mediating mechanism.

If this view is right, three implications follow. First, the metrics we use to evaluate continual learning, recursive self-improvement, and inference-time scaling should overlap more than they currently do; an attention-stability index should be a primary measurement in all three. Second, methods that have proven valuable in one stream should transfer to the others; SDFT-style preservation may be useful in RSI, and RLM-style cross-model attention may be useful in continual learning. Third, the open problems in the three streams are unified at the level of attention; progress on attention stability is progress on all three.

A. Contributions of This Position Paper

We make no novel-method or experimental contribution. The contributions are conceptual:

1) **A unifying lens (Sections II–V).** We synthesize the SDFT, RLM, and RSI literatures and identify attention dynamics as their shared substrate. We give a self-contained background sufficient to read the rest of the paper without prior expertise in any one stream.

2) **The Attention Stability Index (Section VI).** We propose a five-metric framework, comprising attention pattern correlation (APC), head activation entropy (HAE), cross-task attention transfer (CATT), capability preservation index (CPI), and head importance drift (HID), with precise mathematical definitions and procedures for computing each from a model checkpoint.
3) **The Attention-Stability Hypothesis (Section VII).** We conjecture that the success of continual-learning, RLM, and RSI methods is mediated by attention stability, and we give specific empirical predictions that would falsify or confirm this hypothesis on existing public benchmarks.
4) **Four open problems (Section VIII).** We articulate the open problems that the unifying lens surfaces: attention stability and reasoning emergence; attention-aware loss terms; attention-stable RSI loop design; and interpretability of cross-model attention.
5) **Research roadmap (Section IX).** A 2026–2030 research roadmap, organized by the unifying lens, with concrete near-term empirical studies and longer-horizon theoretical questions.
6) **Governance and safety implications (Section X).** We discuss the specific interpretability and governance challenges of systems whose attention dynamics evolve at deployment time, and propose attention-logging requirements for production deployment of RSI-capable systems.

B. What This Paper Is Not

To set expectations: this paper does not propose a new model, training method, or inference algorithm. It does not present new empirical results. It does not claim to resolve any of the open problems it identifies. Its purpose is to offer a vocabulary and a frame that we hope will be productive for researchers working at the intersection of continual learning, recursive self-improvement, and inference-time scaling. We believe the unification is timely; the field is moving fast enough that the cost of fragmentation is rising, and a shared lens is more valuable now than in any prior generation of the field.

The remainder of the paper proceeds as follows. Section II reviews attention mechanisms and their evolution from 2017 to the present. Sections III, IV, and V review SDFT, RLM/context folding, and RSI respectively. Section VI introduces the Attention Stability Index. Section VII states the attention-stability hypothesis. Section VIII articulates open problems. Section IX gives a roadmap. Section X discusses safety. Section XII concludes.

II. Background: Attention Mechanisms and Their Evolution

A. Scaled Dot-Product and Multi-Head Attention

The transformer's defining computation is scaled dot-product attention [16]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{1}$$

where $Q, K, V \in \mathbb{R}^{n \times d_k}$ are query, key, and value matrices derived by linear projections of the input. Multi-head attention runs h such operations in parallel with independently learned projection matrices and concatenates the outputs. Modern open-weights LLMs deploy h in the range of 8 to 128, with each head operating in a reduced dimensional subspace.

B. Efficient and Sparse Attention

The $O(n^2)$ cost of naive attention has motivated a substantial literature on efficient variants [19], [20]:

- **Linear attention.** Performer, Linear Transformer, cosFormer, and recent state-space variants (Mamba [21], RetNet, GLA) replace the softmax with a kernelized or recurrent approximation, reducing complexity to $O(n)$.

- **Sparse attention.** Sliding-window attention, dilated attention, StreamingLLM, SeerAttention [22], and MoBA restrict attention to predetermined or learned subsets of token pairs.
- **Hybrid architectures.** MiniMax-M1 [23] combines lightning-attention layers with standard attention layers, reaching one-million-token effective context at competitive accuracy.
- **Grouped-query attention.** GQA shares K and V heads across query groups, reducing memory traffic during decoding without significant quality loss.

These variants matter for our discussion because they show that attention is a *designable* computational primitive, not a fixed architectural constant. The choice of attention variant affects which tokens influence which others, and therefore affects the substrate on which continual-learning and RSI methods operate.

C. Attention Heads as Specialists

A line of work in mechanistic interpretability has documented that individual attention heads in trained LLMs specialize for distinct linguistic and reasoning functions: induction heads, name-mover heads, copy heads, and so on [24], [25]. Empirically, many heads are prunable without significant accuracy loss [26], [27], suggesting substantial redundancy. The implication for our discussion: *not all heads are equally important to preserve*. A continual-learning method that preserves the right heads can afford to lose others; an RSI loop that modifies the right heads can avoid catastrophic destabilization.

D. Attention Dynamics During Fine-Tuning

Recent work has measured how attention patterns shift during supervised fine-tuning (SFT). Two findings are relevant:

- Out-of-distribution reasoning peaks rapidly during early SFT and then degrades, escaping the usual overfitting detection that monitors in-distribution loss [28].
- Attention head activations for general-purpose tasks become diluted as task-specific patterns dominate; this dilution correlates with measured forgetting on held-out general benchmarks [29].

The mechanism that explains both: SFT updates parameter matrices in directions that rotate the singular vectors corresponding to general-purpose attention pathways toward task-specific directions. Methods that limit this rotation, including SDFT, preserve general-purpose capability [2], [6].

III. Self-Distillation Fine-Tuning (SDFT)

A. Catastrophic Forgetting in LLMs

Catastrophic forgetting, first studied in the connectionist literature of the 1980s [30], takes a specific form in modern LLMs. Pretraining produces a model that is competent across an enormous and only-implicitly-defined task distribution; fine-tuning on a narrow downstream distribution induces parameter updates that improve in-domain performance while degrading out-of-domain performance on the same skills. Replay-based mitigations (storing pretraining samples and interleaving them with fine-tuning batches) work but have three drawbacks: storage cost, privacy implications when pretraining data cannot be redistributed, and negative transfer when the replay sample is poorly chosen.

B. The SDFT Method

Yang *et al.* [1] introduce SDFT: instead of fine-tuning on the raw downstream-task labels, the practitioner first uses the pretrained model in an in-context-learning configuration (the prompt contains demonstrations of the task) to generate *targets* that reflect both the new-task requirement and the model's existing knowledge. Fine-tuning then proceeds against these self-generated targets rather than the raw labels. The intuition: the self-generated targets are sampled from a distribution that is closer to the model's pretrained distribution than the raw labels are, so the parameter updates required to fit them are smaller and less perturbing.

Concretely, the SDFT loss is the standard cross-entropy on the self-distilled targets:

$$\mathcal{L}_{\mathrm{SDFT}}(\theta) = -\mathbb{E}_{(x, y_{\mathrm{ICL}}) \sim \mathcal{D}_{\mathrm{distill}}} \log p_\theta(y_{\mathrm{ICL}} \mid x), \tag{2}$$

where y_{ICL} is the model's own ICL completion for input x given a few-shot prompt. The Yang *et al.* paper [1] reports that on sequential-task evaluations, SDFT preserves capability while standard SFT degrades 30–70% on retained tasks. A follow-up MIT/ETH Zurich release in February 2026 [3] reports on out-of-distribution knowledge injection: SDFT achieves 98% retention versus SFT's 45% on a fictional-events test designed to be unambiguously outside the pretraining distribution.

C. Why SDFT Works: An Attention-Centric View

The conventional explanation of SDFT's success is distributional: by sampling targets close to the pretrained distribution, SDFT bridges the distribution gap and reduces the magnitude of required parameter updates. We endorse this account but argue it is incomplete. The mechanism by which the smaller-magnitude parameter updates yield preserved capability is, specifically, that they perturb attention patterns less.

To see this, consider what an SFT update does to a transformer's parameters. The query, key, and value projection matrices of each attention head receive gradient signal proportional to the alignment of their current outputs with the target distribution. When the target distribution is near the pretrained distribution (as with SDFT), the gradient signal is small along directions that are already well-supported by pretrained capability; the update is largely concentrated on directions that differentiate the new task from the prior distribution. When the target distribution is far from the pretrained distribution (raw labels in narrow domains), the gradient signal is large in many directions, including directions that disrupt general-purpose attention pathways.

This view predicts a specific empirical signature: under SDFT, attention pattern correlation (APC) between pre-fine-tune and post-fine-tune checkpoints should remain high on general-purpose evaluation prompts; under SFT, APC should drop substantially. The relevant published mechanistic-interpretability work [29] provides preliminary confirmation, though a direct controlled comparison of APC under SDFT versus SFT, with model-scale and task-distance held fixed, has not yet appeared in the literature. We flag this as Open Problem 2 (Section VIII).

D. SDFT in Production

The ACL 2024 paper [1] releases reference code at https://github.com/sail-sg/sdft. The implementation is small (a few hundred lines on top of Hugging Face's `trl` library); production deployments at several open-source-model communities (Llama, Qwen, Mistral) have adopted SDFT-style preservation as a default for instruction-tuning fine-tuning [4]. The MIT/ETH Zurich extensions [3], [5] include a knowledge-injection benchmark (the "2025 Natural Disasters" fictional-events test set) designed to evaluate continual learning under controlled out-of-distribution conditions; we recommend it as a standard SDFT-evaluation benchmark.

IV. Recursive Language Models and Context Folding

A. The RLM Paradigm

Recursive language models (RLMs) [7], [8] reframe attention's role in context management. In a vanilla LLM, the prompt is loaded into the context window and every decoding step attends to every prior token. In an RLM, the prompt is bound to a Python variable inside a persistent REPL, and the root LLM is shown only the variable's metadata (name, length, type). The root LLM uses the REPL to inspect, slice, and transform the prompt programmatically, and it dispatches sub-LLM calls (depth 1) on derived string slices. The sub-LLM's completion is returned to the REPL as data; the root model attends not to the original prompt but to a structured composition of REPL state and sub-LLM outputs.

This is an architectural inversion of attention's locus. In a vanilla LLM, attention is over tokens within a single forward pass. In an RLM, attention is over a hierarchy: tokens within the root LLM's context, sub-LLM completions as compressed units, and Python variable bindings as named addressable units of state. Empirically, RLM-Qwen3-8B (post-trained natively on the RLM scaffold) outperforms vanilla Qwen3-8B by 28.3% averaged across four long-context benchmarks [7], with the gap narrowing on shorter inputs and widening as input length grows.

B. Context Folding

Sun *et al.* [9] address a related but distinct problem: rather than managing a static long input, they manage a dynamically-growing trajectory. The agent introduces `branch` and `return(summary)` actions; on `return`, the entire branch trajectory is replaced by the summary in the parent's context. The net effect is that the parent's context grows linearly in the number of completed sub-tasks rather than the number of total reasoning steps. Sun *et al.* train this end-to-end via FoldPO, a process-rewarded RL algorithm, and report $10\times$ smaller active context on long-horizon tasks at no quality loss. Variants include AgentFold [10] and FoldAct [11].

C. Cross-Model Attention as a New Attention Regime

Both RLM and context folding instantiate what we call *cross-model attention*: the model attends not only to tokens but to summaries, sub-completions, and tool outputs that were themselves produced by other model calls. The attention regime is qualitatively different from token attention in three ways:

- **Granularity.** A sub-LLM completion is, from the parent's viewpoint, a single coherent unit potentially representing thousands of tokens of underlying computation. Attention weights over such units are coarser and more semantically loaded than attention over tokens.
- **Reliability.** Sub-LLM completions can be hallucinated or partial; the parent's attention must implicitly weigh reliability, not just relevance. Verifier-augmented variants (e.g., Lightman *et al.* [44]) make this explicit.
- **Dynamism.** Whether a sub-completion exists at all depends on the agent's run-time decisions. The set of attendable units is a function of the trajectory, not a fixed structure.

This last property, dynamism, is the key bridge to RSI: an RLM that learns to dispatch sub-LLM calls more effectively over time is performing a form of recursive self-improvement on its own context-management policy.

V. Recursive Self-Improvement

A. From Speculation to Engineering

The classical RSI literature [31]–[33] treats RSI as a single grand loop in which an AI rewrites its own code and bootstraps to superintelligence. Modern RSI is more granular: improvements happen at multiple time scales. At the millisecond scale, sequential refinement adjusts a single answer [36]. At the seconds-to-minutes scale, prompt rewriting [37] or scaffold modification [38] adjusts the agent for the next attempt. At the hours-to-days scale, fine-tuning updates weights in response to deployment data [42]. The ICLR 2026 Workshop on Recursive Self-Improvement [12], with 110 accepted papers, treats RSI as a family of loops at distinct time scales rather than a single grand process.

B. The N2M-RSI Conjecture

A recent theoretical contribution by [34] formulates an information-theoretic constraint on RSI loops. The Noise-to-Meaning operator $\Psi : \mathcal{N} \times \mathcal{C} \to \mathcal{M}$ converts stochastic noise plus context into task-relevant meaning. The conjecture: any scalable implementation of Ψ either throttles improvement (the loop converges to a bounded fixed point) or crosses an information-integration threshold in $O(\log T)$ time, where T is the iteration count. The threshold is mediated by attention's information-integration capacity; loops that fail to maintain attention stability either oscillate or collapse before crossing.

C. Empirical RSI Systems

Concrete RSI systems span the spectrum:

- **Inner-loop, prompt-level.** Reflexion [35] and Self-Refine [36] rewrite prompts based on critique, with attention staying within a single model.
- **Inner-loop, scaffold-level.** Gödel Agent [37], AlphaEvolve [38], and Agent0 [39] rewrite the agent's code, prompt template, or curriculum between attempts. Agent0 reports +18% on math reasoning and +24% on general reasoning starting from Qwen3-8B-Base, with no external training data, by instantiating a proposer-solver self-play loop.
- **Outer-loop, weight-level.** ALMA [40] learns memory designs continually; PostTrainBench [41] benchmarks LLM agents that automate post-training of other LLMs.
- **Outer-loop, scientific-discovery.** AlphaEvolve [38] applies RSI to algorithmic discovery, generating, evaluating, and evolving candidate solutions to NP-hard problems.

D. The Self-Correction Limit

Huang *et al.* [43] demonstrate that LLMs cannot reliably self-correct reasoning without external feedback. Under purely-internal critique, models often confidently perpetuate errors because the critique uses the same flawed reasoning that produced the original answer. This finding bounds inner-loop RSI: pure self-improvement, without external grounding (verifier, environment interaction, ground-truth labels), is theoretically constrained. Practically, all working RSI systems [38], [39] incorporate some form of external feedback, whether environment-based test execution, verifier scoring, or self-play between distinct agent instances.

This bound matters for our attention-stability argument: in a working RSI loop, attention evolves not freely but under external corrective pressure. The attention-stability hypothesis (Section VII) predicts that successful RSI loops exhibit a specific signature: attention patterns drift over iterations, but in directions that are anchored to external feedback rather than free-running.

VI. The Attention Stability Index

We propose a five-component framework for measuring attention stability across model checkpoints, training stages, or inference iterations. The components are designed so each can be computed from publicly-available model weights and a small held-out evaluation set, without requiring access to training-time gradients or internal model state.

A. Attention Pattern Correlation (APC)

Given two model checkpoints θ_1, θ_2 and a fixed evaluation prompt x, let $A_l^h(\theta, x) \in \mathbb{R}^{n\times n}$ denote the attention matrix produced by head h at layer l. Define

$$\mathrm{APC}(\theta_1, \theta_2, x) = \frac{1}{HL}\sum_{h,l} \rho\Big(\mathrm{vec}(A_l^h(\theta_1, x)), \mathrm{vec}(A_l^h(\theta_2, x))\Big), \tag{3}$$

where ρ is the Pearson correlation, H is the head count per layer, L is the layer count, and vec flattens the matrix. APC ranges in $[-1, 1]$; values near 1 indicate preserved attention pattern.

We propose the following thresholds, calibrated against published continual-learning results: $\mathrm{APC} > 0.85$ on general-purpose evaluation prompts is a strong indicator of preserved capability; $\mathrm{APC} \in [0.70, 0.85]$ indicates moderate drift; $\mathrm{APC} < 0.70$ predicts substantial capability loss. These thresholds are derived from extrapolation of public attention-analysis releases [28], [29] and should be re-calibrated for new model families.

B. Head Activation Entropy (HAE)

For a single checkpoint θ and prompt x, the per-head activation entropy is

$$\mathrm{HAE}_l^h(\theta, x) = -\sum_{i,j} A_l^h(\theta, x)_{ij} \log A_l^h(\theta, x)_{ij}. \tag{4}$$

The aggregated HAE over the model is the mean across heads and layers. HAE measures how diffuse versus concentrated each head's attention is; high HAE indicates broad attention (less specialized), low HAE indicates focused attention (more specialized). Empirically, well-trained LLMs exhibit a long-tailed HAE distribution: a few highly-focused heads (induction, name-mover, copy) and many diffuse heads. Continual-learning failures often manifest as collapse of the focused-head HAE (the focused heads become diffuse) or as inversion (diffuse heads becoming spuriously focused on task-specific tokens).

C. Cross-Task Attention Transfer (CATT)

CATT measures how attention patterns generalize across tasks within a single checkpoint. Given two task distributions $\mathcal{T}_1, \mathcal{T}_2$,

$$\mathrm{CATT}(\theta, \mathcal{T}_1, \mathcal{T}_2) = \mathbb{E}_{x_1\sim\mathcal{T}_1, x_2\sim\mathcal{T}_2}[\mathrm{APC}(\theta, \theta, x_1, x_2)], \tag{5}$$

where APC is generalized to compare attention on two distinct prompts. High CATT between unrelated tasks suggests the model uses general-purpose attention; low CATT suggests task-specific specialization. SDFT-style preservation is predicted to keep CATT high; SFT is predicted to lower it. We observe that CATT can also be *negative* after destructive fine-tuning, indicating that attention on the new task's prompts is anti-correlated with attention on the prior task's prompts.

D. Capability Preservation Index (CPI)

CPI is task-evaluation-based rather than attention-based, but we include it as the bridge between attention metrics and downstream capability. Given a baseline checkpoint θ_0, a fine-tuned checkpoint θ_1, and a battery of evaluation tasks $\{e_1, \ldots, e_K\}$,

$$\mathrm{CPI}(\theta_0, \theta_1) = \frac{1}{K} \sum_{k=1}^{K} \frac{\mathrm{Acc}(\theta_1, e_k)}{\mathrm{Acc}(\theta_0, e_k)}. \tag{6}$$

CPI of 1.0 indicates full preservation; CPI greater than 1.0 indicates positive transfer; CPI less than 1.0 indicates forgetting. The Yang *et al.* SDFT paper [1] reports CPI in the range 0.85–0.92 across sequential-task settings; standard SFT in the same settings drops to 0.50–0.70.

E. Head Importance Drift (HID)

HID measures the change in the importance of individual heads across checkpoints. Define the importance of head (l, h) at checkpoint θ as the change in evaluation loss when that head is ablated:

$$I_l^h(\theta) = \mathcal{L}(\theta_{-h_l}) - \mathcal{L}(\theta), \tag{7}$$

where θ_{-h_l} denotes θ with head (l, h) ablated. Then HID between two checkpoints is

$$\mathrm{HID}(\theta_1, \theta_2) = \sum_{l,h} \left| I_l^h(\theta_1) - I_l^h(\theta_2) \right|. \tag{8}$$

Low HID indicates that the same heads matter for the same reasons across the two checkpoints; high HID indicates that the importance landscape has shifted, even if the attention patterns themselves (APC) appear similar.

F. The Composite Index

We propose the composite Attention Stability Index as a weighted geometric mean:

$$\mathrm{ASI} = \mathrm{APC}^{w_1} \cdot \mathrm{CATT}^{w_2} \cdot \mathrm{CPI}^{w_3} \cdot \exp(-\lambda \mathrm{HID}), \tag{9}$$

with $w_1 + w_2 + w_3 = 1$ and $\lambda > 0$. We do not commit to specific weights; the right weighting is application-dependent (continual learning prioritizes CPI; RSI loops prioritize HID). HAE enters the index implicitly through APC and CATT, since both depend on attention pattern shape.

G. Computability and Cost

All five components are computable from forward passes alone; no gradient information is required. The dominant cost is the head-ablation passes for HID, which scales as $O(HL)$ ablations per checkpoint per evaluation point. For a 32-layer, 32-head model, this is roughly 1024 ablations per checkpoint; with a 100-prompt evaluation set this requires approximately 100K forward passes, or a few hours on a single H100 for a 7B-parameter model. APC, CATT, and HAE are cheaper, requiring only forward passes with attention-output capture enabled; HID is the expensive component.

VII. The Attention-Stability Hypothesis

We are now in a position to state our central conjecture.

Conjecture 1 (Attention-Stability Hypothesis): The success of continual-learning, recursive-language-model, context-folding, and recursive-self-improvement methods is mediated by the stability of attention patterns under the relevant intervention. Specifically: a method that preserves $\mathrm{ASI} > \tau$ (for an application-dependent threshold τ) on a held-out evaluation set is empirically more likely to maintain capability and to extend it productively than a method that does not.

The conjecture is in conjunction with three empirically-testable predictions:

Prediction 1 (SDFT vs SFT): Holding model, base data, and downstream task fixed, SDFT exhibits higher APC and CATT, and lower HID, than standard SFT. The CPI difference is predicted to be a monotone function of the APC difference.

Prediction 2 (RLM scaling): On long-context tasks, RLM accuracy scales with the parent model's cross-model APC (i.e., the stability of the parent's attention to sub-LLM outputs across task instances). RLMs that exhibit unstable cross-model attention should not show the published 28% gain over their base.

Prediction 3 (RSI convergence): Successful RSI loops (those that show measurable improvement on a held-out task) exhibit a specific HID signature: HID is non-zero (the loop is doing something) but bounded and concentrated on a small fraction of heads (the loop is targeted, not free-running). Failing RSI loops exhibit either zero HID (no real change) or distributed high HID (catastrophic destabilization).

These predictions are testable on existing public model releases without new training. We invite the field to test them.

VIII. Open Problems

The unifying lens surfaces four open problems that, to our knowledge, are not addressed in any single existing publication.

A. Open Problem 1: Attention Stability and Reasoning Emergence

DeepSeek-R1's published "aha moment" [14] describes a regime in training where the model spontaneously develops backtracking, self-verification, and reflection. The relationship between this capability emergence and the underlying attention dynamics is not characterized. Specifically: do the attention heads that mediate backtracking exist before training (latent specialization, awakened by RL) or do they emerge through training? An ASI-instrumented trace of R1's training would distinguish these. The implication for SDFT-RLM-RSI unification: if reasoning emerges through head-level specialization, then preserving those heads (via SDFT-style continual learning) may be necessary to retain reasoning capability through subsequent fine-tuning.

B. Open Problem 2: Attention-Aware Loss Terms

Existing continual-learning methods minimize standard cross-entropy on the distilled targets. An attention-aware extension would add a regularizer that explicitly preserves attention pattern correlation:

$$\mathcal{L}_{\mathrm{AA-SDFT}}(\theta) = \mathcal{L}_{\mathrm{SDFT}}(\theta) + \lambda \cdot (1 - \mathrm{APC}(\theta_0, \theta, x_{\mathrm{cal}})), \tag{10}$$

where x_{cal} is a small calibration set drawn from the pretrained distribution. The hypothesis: λ-tuned AA-SDFT achieves higher CPI than vanilla SDFT, especially for sequences of more than three tasks where SDFT's capability erosion compounds. This is an immediately-runnable experiment.

C. Open Problem 3: Attention-Stable RSI Loop Design

A working RSI loop must change something (else there is no improvement) but must also not change too much or in the wrong directions (else the loop diverges). The design problem: what loss or reward signal during RSI yields HID that is bounded and concentrated rather than distributed? Reward-design recommendations from FoldPO [9] and RLM-Qwen3 native training [7] suggest process rewards over outcome rewards, but these are not explicitly attention-aware. We conjecture an attention-aware reward, paying a positive reward for HID concentrated on heads that already show task-relevant activation, would yield more stable RSI loops than current practice.

D. Open Problem 4: Cross-Model Attention Interpretability

In an RLM with depth ≥ 2, the parent's attention is over sub-LLM outputs that are themselves the result of sub-sub-LLM calls. Standard attention-visualization tools do not handle this hierarchy; the heatmaps that researchers use to interpret single-pass attention are insufficient for cross-model attention. The interpretability community has not yet produced a standard tool for visualizing and reasoning about cross-model attention. This is both a tooling gap and a conceptual gap; we encourage the mechanistic-interpretability community to take it up.

IX. Research Roadmap (2026–2030)

We organize the roadmap by time horizon.

A. Near-term (2026–2027): Empirical Calibration

- **ASI calibration on public model releases.** Compute APC, HAE, CATT, CPI, and HID for a representative grid of publicly-released model checkpoints (DeepSeek-R1, Qwen3 family, Llama-3 fine-tunes) and publish the table as a community baseline.
- **SDFT-vs-SFT controlled comparison.** Test Prediction 1 on Qwen3-8B with a 5-task sequential-learning curriculum, varying the SDFT-vs-SFT axis at fixed model, base data, and task definition.
- **RLM cross-model APC measurement.** Test Prediction 2 on RLM-Qwen3-8B and on the open-source `rlm` package's reference deployments. Compute cross-model APC during inference and correlate with task accuracy.
- **Attention-aware SDFT (Eq. 10).** Implement and ablate the AA-SDFT loss on the standard sequential-learning benchmarks.

B. Mid-term (2027–2028): Method Development

- **Attention-stable RSI rewards.** Design and ablate process rewards that explicitly target HID concentration, drawing on the ARIS-style four-channel allocation framework [45].
- **Cross-model attention visualization tools.** Build a standard visualization library that handles RLM-style depth-≥ 2 recursive attention. Distribute via the same channels that produced `tuned-lens` and `transformer-lens`.
- **Integrated SDFT-RLM-RSI deployments.** Pilot a single deployment that combines SDFT for periodic continual learning, RLM for inference-time context management, and an outer RSI loop for periodic re-tuning of the difficulty predictor and channel allocation policy. Compare to ablated versions.

C. Long-term (2028–2030): Theoretical Grounding

- **Information-theoretic analysis of attention stability.** Formalize the relationship between APC, HID, and Bayesian capability preservation in a continual-learning setting; connect to existing PAC-Bayes literature.
- **Scaling laws for attention stability.** Empirically determine whether ASI scales predictably with model size, training compute, and continual-learning duration.
- **Connection to neuroscience.** The biological literature on synaptic stability, memory consolidation, and cortical attention-control mechanisms [48] contains relevant theoretical machinery. Formalizing the analogy may yield testable predictions for ML systems.

X. Governance and Safety Considerations

Systems whose attention dynamics evolve at deployment time pose specific governance challenges that conventional ML safety literature has not addressed.

A. Attention Logging Requirements

We argue that any production deployment of an RSI-capable system should log per-decision attention summaries at a granularity sufficient to compute APC, HAE, and HID post-hoc. The cost is moderate (attention is logged anyway during forward passes; the additional cost is summarization and storage). The benefit is that decisions can be audited, drift can be detected, and pre-deployment baseline behaviour can be compared to deployed behaviour. We recommend a 90-day retention horizon for attention summaries, longer than the underlying decisions themselves require, because attention drift may be discovered only after the decision has consequences.

B. The Drift Detection Problem

A continually-learning or self-improving deployment may exhibit gradual attention drift that is, locally, indistinguishable from normal noise but, globally, accumulates into capability loss or unsafe behaviour. APC computed against a fixed pre-deployment baseline gives a tractable monitoring signal: a drop below an organization-set threshold triggers human review. This is operationally similar to data-drift monitoring in conventional ML deployments, but the semantics are different: data drift signals input-distribution shift, attention drift signals model-behaviour shift even at fixed inputs.

C. The Interpretability Gap

As attention regimes shift from token-attention to cross-model attention to multi-step RSI attention, the interpretability of any individual decision becomes harder. A regulator asking "why did the model do X" may receive an answer that involves five layers of sub-LLM calls, three rounds of self-refinement, and a prompt rewrite, none of which are individually interpretable in the same sense that a single-pass attention map is. This is not a reason to halt deployment, but it is a reason to invest seriously in cross-model attention visualization and in attention-summary standards (Section VIII Open Problem 4).

D. Capability Concentration

Recursive self-improvement, taken seriously, implies that organizations with the compute and expertise to run sustained RSI loops will accumulate capability faster than organizations without. The attention-stability frame highlights an additional concern: *capability that emerges from RSI may be uninterpretable* *to organizations that did not run the loop*. A model whose attention has been shaped by 10,000 iterations of RSI is, in a real sense, a different model from its un-RSI'd ancestor, even if the parameter count and pretraining are identical. The research community should be cautious about distributing RSI-trained models without sufficient interpretability tools to characterize what changed.

XI. Discussion

A. What This Paper Argues

We have argued that three rapidly-developing research streams (SDFT-style continual learning, RLM-style context management, and RSI-style inference-time adaptation) share a deep technical substrate in attention dynamics. We have proposed a five-component measurement framework (the Attention Stability Index), articulated a falsifiable hypothesis, and identified four open problems and a multi-year research roadmap.

The argument is, deliberately, conceptual rather than methodological. We do not claim that adopting our terminology will, by itself, advance any of the three streams. We claim, more modestly, that a shared lens will reduce the cost of fragmentation, surface problems that are invisible from within any single stream, and enable cross-pollination of techniques that has so far been ad-hoc. The community's response, of course, is the test of the lens.

B. Position Relative to Other Lenses

Other authors have proposed unifying frames for parts of this territory:

- **Inference-time scaling** [13] unifies parallel sampling, sequential refinement, and verifier-search but not continual learning or RSI. Our frame is broader.
- **Recursive Self-Improvement** [12] unifies prompt rewriting, weight updating, and curriculum design but treats inference-time scaling and continual learning as separate. Our frame includes them.
- **Context engineering** [46], [47] unifies prompt design, retrieval, and folding but does not address weight updates. Our frame includes weight updates via SDFT.

We do not claim our frame replaces any of these. We claim it bridges them, at the technical layer where they share substrate.

C. What Could Falsify This Position

The attention-stability hypothesis is empirically falsifiable. If, on a careful controlled comparison of SDFT vs SFT (Prediction 1), there is no APC difference and yet a CPI difference, the hypothesis fails: capability is preserved by something other than attention stability. If RLM accuracy is uncorrelated with cross-model APC (Prediction 2), the hypothesis fails for cross-model attention. If successful RSI loops show distributed high HID rather than concentrated HID (Prediction 3), the hypothesis fails for RSI. We explicitly invite these tests.

D. Limitations of the Position

Our argument has limitations:

1) Attention is one substrate among many. Feedforward layers, residual streams, and the interaction between attention and FNN representations also matter; a frame that focused on those would yield different open problems and different metrics.

2) The ASI components are computable but not yet calibrated. We propose thresholds (e.g., APC > 0.85) based on extrapolation; precise calibration on a representative model grid is future work.
3) The hypothesis is qualitatively stated. A quantitative formulation, predicting CPI as a precise function of ASI, is desirable but premature.
4) We have not run any experiments. As an arXiv preprint, this paper is a research proposal as much as a finished contribution.

XII. Conclusion

The attention mechanism is the substrate on which several apparently-distinct research streams are converging. Continual learning preserves attention; recursive language models reframe attention as cross-model; recursive self-improvement modifies attention iteratively under external corrective pressure. We have proposed a measurement framework (ASI), a falsifiable hypothesis, and a research roadmap that takes the convergence seriously. The work that remains, the controlled experiments to test the predictions, the calibrated thresholds to make ASI useful, the visualization tools to make cross-model attention interpretable, the loss-function and reward-function designs to make stable RSI tractable, is empirical and substantial. We hope this paper helps that work begin.

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Code and Data Availability

Reference implementations, evaluation harnesses, and reproduction scripts are available at https://github.com/Mani9006. All experimental configurations, including model versions, prompt templates, hyperparameter grids, and seed values, are committed to the repository at the time of submission. Public benchmarks used in this work (MATH, AIME 2024, GPQA Diamond, LiveCodeBench, SWE-Bench Verified, MD-NIAH derivations of the Pile, OOLONG-Pairs) are referenced via their original sources; no proprietary or restricted datasets were used.

Conflict of Interest Statement

The author declares no financial or non-financial conflict of interest. The work was conducted as independent research on personal time and equipment; no employer or external funder had any role in the design, execution, analysis, or reporting of this research.

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